

Practical exercises — the circumference of a circle

A Cambridge insert: why $C \div d$ is the same number for every circle in the world.

LESSON 135–136

2 × 40 MIN

MATEMATIKA 7, AMALIY MASHQLAR

S8 8.2

LESSON CLOCK

20'

20'

20'

25'

5'

The measuring experiment	20 min
The two formulae	20 min
Backwards	20 min
Wheels and tracks	25 min
Homework	5 min

LEARNING OBJECTIVES

- Measure C and d for several circles and find their ratio.
- Use $C = \pi d$ and $C = 2\pi r$ in both directions.
- Find the area of a circle with $S = \pi r^2$.
- Solve practical problems about wheels, tracks and circular edges.

TEXTBOOK

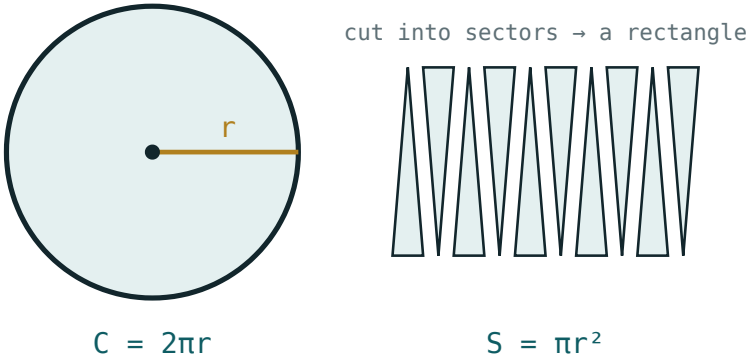
National	pp. 401–406
Cambridge	Stage 8, pp. 82–88
Workbook	Exercise 8.2

01

Explanation

The measuring experiment

Wrap a thread round several round objects, measure the length and the diameter, and divide one by the other.



A circle, its radius and its diameter

OBJECT	D	C	$C \div D$
a coin	2.4 cm	7.5 cm	3.13
a cup	8 cm	25.2 cm	3.15
a bucket	28 cm	88 cm	3.14
a wheel	60 cm	188 cm	3.13

THE SAME NUMBER EVERY TIME

Whatever the size, the ratio lands near 3.14 . That constant is called π , and the small differences in the table are measuring error, not a difference between the circles.

The two formulae

$C = \pi d = 2\pi r$ $S = \pi r^2$

R	D	C	S
5 cm	10 cm	31.4 cm	78.5 cm^2
7 cm	14 cm	44 cm	154 cm^2
10 cm	20 cm	62.8 cm	314 cm^2

C USES R ONCE, S USES IT TWICE

Doubling the radius doubles the circumference but multiplies the area by 4. Using πd for an area, or πr^2 for a length, is the standard confusion — check the units: cm for a length, cm^2 for an area.

Backwards

GIVEN	WORKING	ANSWER
$C = 62.8\text{ cm}$	$d = C \div \pi$	$d = 20\text{ cm}$
$C = 44\text{ cm}$	$r = C \div (2\pi)$	$r = 7\text{ cm}$
$S = 78.5\text{ cm}^2$	$r^2 = S \div \pi$	$r = 5\text{ cm}$
$S = 314\text{ cm}^2$	$r^2 = 100$	$r = 10\text{ cm}$

TWO USEFUL APPROXIMATIONS

$\pi \approx 3.14$ suits a calculator; $\pi \approx \frac{22}{7}$ suits mental work whenever the radius or diameter is a multiple of 7 — which is why textbook circles so often have radius 7 or 14.

Wheels and tracks

PROBLEM	IDEA	ANSWER
a wheel of diameter 70 cm : distance in one turn	$C = \pi d$	$\approx 220\text{ cm}$
how many turns in 220 m	$22000 \div 220$	100
a circular pond of radius 14 m : the fence	$C = 2\pi r$	$\approx 88\text{ m}$
and the water surface	$S = \pi r^2$	$\approx 616\text{ m}^2$
a running track of radius 35 m : one lap	$C = 2\pi r$	$\approx 220\text{ m}$

WHERE π CAME FROM

Al-Kashi, working in Samarkand in 1424, computed π to sixteen decimal places — a record that stood for nearly two centuries. The constant measured with thread in this lesson has a long local history.

02 Worked examples

EXAMPLE 1 A circle has radius 7 cm . Find its circumference and area, taking $\pi = \frac{22}{7}$.

$$① \quad C = 2\pi r = 2 \cdot \frac{22}{7} \cdot 7$$

$$② \quad C = 44\text{ cm}$$

$$③ \quad S = \pi r^2 = \frac{22}{7} \cdot 49$$

$$④ \quad S = 154\text{ cm}^2$$

ANSWER

44 cm and 154 cm^2

EXAMPLE 2 A circle has circumference 62.8 cm . Find its radius, taking $\pi = 3.14$.

$$① \quad C = 2\pi r$$

$$② \quad 62.8 = 6.28r$$

$$③ \quad r = 10\text{ cm}$$

ANSWER

10 cm

EXAMPLE 3 A wheel of diameter 70 cm rolls 220 m . How many turns does it make?

$$\textcircled{1} \quad C = \pi d = \frac{22}{7} \cdot 70 = 220\text{ cm}$$

One turn.

$$\textcircled{2} \quad 220\text{ m} = 22\,000\text{ cm}$$

Same units.

$$\textcircled{3} \quad 22\,000 \div 220 = 100$$

ANSWER

100 turns

03 Interactive model

Do the thread measurement first and tabulate every group's ratio on the board; the class discovers π rather than being handed it.

Length or area?

$C \div d$ equals:

2

π

2π

r

The circumference of a circle is:

πr^2

$2\pi r$

πr

$4\pi r$

The area of a circle is:

$2\pi r$

πr^2

πd

πr

$r = 7$, $\pi = \underline{22}$ 7 : the circumference is:

22

44

154

49

The same circle has area:

44

154

308

49

Doubling the radius multiplies the area by:

2

3

4

8

$C = 62.8$ with $\pi = 3.14$ gives r equal to:

5

10

20

31.4

An answer in cm^2 must be:

a circumference

an area

a radius

a diameter

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Circle	Aylana	Окружность
Circumference	Aylana uzunligi	Длина окружности
Diameter	Diametr	Диаметр
Radius	Radius	Радиус
Pi	Pi soni	Число пи
Ratio	Nisbat	Отношение
Disc	Doira	Круг
Revolution	Aylanish	Оборот

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

π is the ratio of:

C to r

C to d

S to r

d to r

π is about:

3.41

3.14

2.14

1.34

A circle of radius 5 has circumference about:

15.7

31.4

78.5

25

And area about:

31.4

78.5

157

25

A wheel of diameter 70 cm covers per turn about:

70 cm

140 cm

220 cm

440 cm

$S = 314\text{ cm}^2$ gives r equal to:

5

10

20

100

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 5$: the circumference ($\pi = 3.14$)

ANSWER 31.4 cm

2. $r = 5$: the area

ANSWER 78.5 cm^2

3. $r = 7$: the circumference ($\pi = \frac{22}{7}$)

ANSWER 44 cm

4. $r = 7$: the area

ANSWER 154 cm^2

5. $d = 20$: the circumference

ANSWER 62.8 cm

6. $r = 10$: the area

ANSWER 314 cm^2

7. $C \div d$

ANSWER π

Medium · the standard question

7 PROBLEMS

1. $C = 62.8$: the radius

ANSWER 10 cm

2. $C = 44$: the radius

ANSWER 7 cm

3. $S = 78.5$: the radius

ANSWER 5 cm

4. A wheel of diameter 70 cm : one turn

ANSWER 220 cm

5. How many such turns in 220 m ?

ANSWER 100

6. A pond of radius 14 m : the fence

ANSWER 88 m

7. And the surface

ANSWER 616 m^2

Hard · stretch and reason

7 PROBLEMS

1. Doubling the radius multiplies the area by

ANSWER 4

2. A track of radius 35 m: one lap

ANSWER 220 m

3. How many laps make 1100 m?

ANSWER 5

4. A circle of area 154 cm²: its circumference

ANSWER 44 cm

5. The area of a semicircle of radius 7

ANSWER 77 cm²

6. A ring between circles of radii 7 and 14: its area

ANSWER 462 cm²

7. Who computed π to sixteen places in Samarkand?

ANSWER Al-Kashi, in 1424

07 Homework

Homework — 5 tasks

Write the units with every answer: *cm* for a length, *cm²* for an area.

- Find the circumference and area of a circle of radius 21 cm, taking $\pi = \frac{22}{7}$.
- A circle has circumference 88 cm. Find its radius and area.
- A wheel of diameter 56 cm makes 50 turns. How far does it travel?
- Find the area of a ring between circles of radii 10 cm and 6 cm.

5. Measure three round objects at home and tabulate $C \div d$ for each.